

Experimental Physics Report

M1 General Physics

Cosmology Data Analysis

Project by

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1 Introduction

Cosmology is the study of the origin and evolution of the universe as a whole. It was a purely philosophical subject in the past, but it has been transformed into an empirical science with high precision within the last 50 years.

The Cosmic Microwave Background (CMB) is the oldest light of the universe, the remnant of the Big Bang dates back to 13.8 billion years ago.

The CMB map was first detected by the COBE satellite ([Smoot et al. 1992](#)). It represents the best-measured thermal black body radiation in nature ([Fixsen 2009](#)), with an average temperature of 2.725 K over all points in the sky. CMB has dipole distortion in temperature due to the motion of the Solar System at about 370 km/s with respect to the CMB. After subtracting the Doppler dipole, the temperature fluctuation is at the scale of 10^{-5} K, proving the fact that the universe is extremely homogeneous and isotropic. The best CMB temperature anisotropies were successfully measured by the European-led space mission Planck, providing the best cosmological model Λ CDM with six parameters measured to extremely high precision. This powerful model describes an expanding universe from an early hot soup of hadrons to the emergence of the complex structures of galaxies and stars we see today.

The inference of cosmological information from the CMB requires us to accurately distinguish the CMB blackbody from other sources: mainly astrophysical sources and instrumental systematic effects. In particular, there are galactic sources like dust, synchrotron and free-free emission, the Sunyaev Zeldovich (SZ) effect, emission from extra-galactic sources, cosmic infrared background, etc. The main framework which has been proven to succeed is component separation; using the different images at different frequencies, we will try to remove every contamination and recover the CMB map. There are two families of component separation: parametric and blind. The first relies on predefined physical models or assumptions about the signal components; notable examples are **FGBuster** and **Commander**. The second makes minimal assumptions about the source signals and separate components based on statistical properties of the data itself; particular successful examples are **SMICA**, **NILC** or **SEVEM**.

This report will be dedicated to reconstructing some of the work that has been done for CMB, in particular with Planck data. We will go briefly through the characteristics of physical processes on the measured spectrum of the HFI instrument of Planck at 6 different frequencies. From 6 maps, we will process them, remove the dipole, smoothing. After that, one of the methods in Internal Linear Combination (ILC) family, the Modified ILC Algorithm (MILCA) is used for component separation. Some extra works on ILC application for Sunyaev Zeldovich effect and parameter constraints from power Spectrum will also be covered.

2 Planck measurements

2.1 Planck Satellite

The Planck satellite, launched in 2009, conducted observations of the entire sky approximately five times over its 2.5-year mission. It was equipped with two main instruments:

- High-Frequency Instrument (HFI): Operated at frequencies of 100, 143, 217, 353, 545, and 857 GHz.
- Low-Frequency Instrument (LFI): Operated at frequencies of 30, 44, and 70 GHz.

These instruments enabled Planck to achieve high-resolution measurements of the cosmic microwave background (CMB) across a broad frequency range.

In this analysis, we will concentrate exclusively on the measurements obtained by the HFI. This choice is motivated by HFI's superior performance, as it provides data with significantly lower noise levels compared to the LFI.

2.2 Point Spread Function

The Point Spread Function describes an imaging system's response to a point source, characterizing the spatial resolution of the instrument. The Full Width at Half Maximum (FWHM) of the PSF is approximately given by:

$$\text{FWHM} \sim \frac{\lambda}{D} \tag{1}$$

where λ is the observed wavelength, and D is the aperture diameter of the telescope. This relationship determines the angular resolution achievable by the system.

2.3 Angular Resolution

Using the HFI frequencies provided in section 2.1 and the FWHM approximation described in section 2.2, we can theoretically estimate the angular resolution. Comparing these theoretical values with the actual FWHM provided by the Planck Collaboration, as shown in figure 1, reveals that the actual FWHM is larger than the theoretical predictions by 1-2 arcminutes. This indicates that the Planck satellite's angular resolution is slightly lower than expected. The reason is likely due to instrumental and systematic effects that were not accounted for in the theoretical approximation. Factors such as imperfections in the telescope optics, alignment errors, diffraction effects, and the finite accuracy of the Planck beam reconstruction process contribute to a broader point spread function.

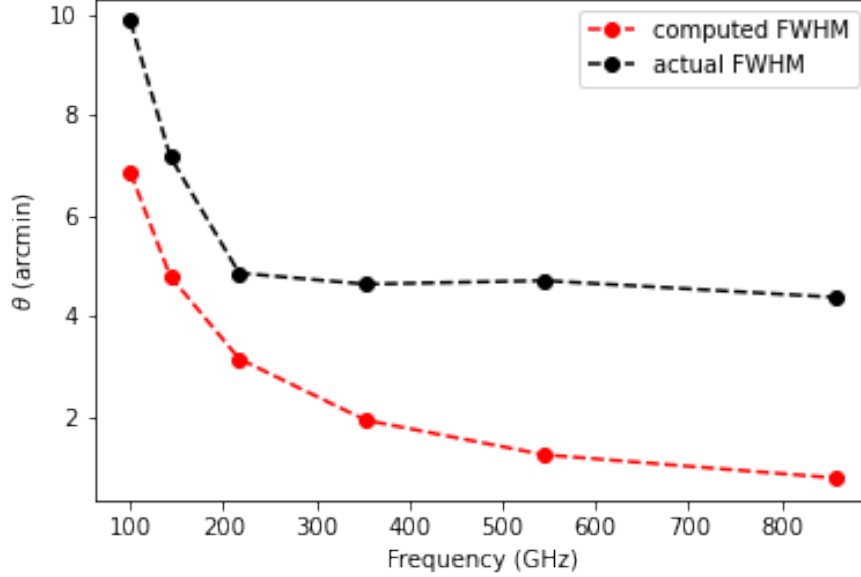


Figure 1: Full-width at half maximum

3 Spectral energy densities (SED)

The CMB map is obscured by many factors, each are plot with the most simple astrophysical model following [Planck Collaboration X \(2016\)](#). The results is shown in figure 2, the models are briefly described as follow:

- **CMB** is a perfect blackbody radiation characterized by Planck's law with one spectral parameter-CMB mean temperature $T_{\text{CMB}} = 2.7255 \pm 0.0006$ K. The CMB model is expressed as:

$$\begin{aligned}
 x &= \frac{h\nu}{k_B T_{\text{CMB}}} \\
 g(\nu) &= (\exp(x) - 1)^2 / (x^2 \exp(x)) \\
 s_{\text{CMB}} &= A_{\text{CMB}} / g(\nu)
 \end{aligned} \tag{2}$$

where h is the Planck constant, ν is the characteristic frequencies of HFI, k_B is the Boltzmann constant, $A_{\text{CMB}} = 70\mu\text{K}$ corresponding to the CMB rms at 1° FWHM resolution,.

- **Synchrotron** is driven by high-energy relativistic cosmic rays such as electrons when they are accelerated by Galactic magnetic field. it is suggested to follow the power law with an index of $\beta_s \sim -3$ at frequencies above 20 Hz:

$$s_s = A_s \left(\frac{\nu}{\nu_0} \right)^{-\beta_s} \tag{3}$$

where $\nu_0 = 408\text{MHz}$ is the reference frequency and $A_s = 30\mu\text{K}$ is the amplitude of the synchrotron radiation spectrum at the reference frequency.

- **Free-free emission** (Bremsstrahlung) occurs when electrons collide with ions, emitting radiation, can be estimated with a power law spectrum at higher frequencies. The model by [Draine \(2011\)](#) shows that the emission measure (EM) scales the emission amplitude, while the electron temperature T_e slightly adjusts the power law index, making it flatter than synchrotron at low frequencies:

$$\begin{aligned} g_{\text{ff}} &= \log \left\{ \exp \left[5.960 - \sqrt{3}/\pi \log \left(\nu_9 T_4^{-3/2} \right) \right] + e \right\} \\ \tau &= 0.05468 T_e^{-3/2} \nu_9^{-2} \text{EM} \ g_{\text{ff}} \\ s_{\text{ff}} &= 10^6 T_e (1 - e^{-\tau}) \end{aligned} \quad (4)$$

where $T_4 = T_e/10^4$, $\nu_9 = \nu/(10^9 \text{ Hz})$, $T_e = 7000 \text{ K}$, $\text{EM} = 100 \text{ cm}^{-3}\text{pc}$

- **Thermal dust** dominant above 100 GHz, arises from the radiation of interstellar dust grains heated by radiations mainly from stars, with a spectrum described by a modified blackbody (graybody) model. The emission is characterized by a free emissivity index β_d and a dust temperature T_d , which vary spatially across the sky. The one-component graybody model can efficiently represent thermal dust up to 857 GHz, including cosmic-infrared background fluctuations that can be interpreted as a weak dust component.

$$\begin{aligned} \gamma &= \frac{h}{k_B T_d} \\ s_d &= A_d \cdot \left(\frac{\nu}{\nu_0} \right)^{\beta_d+1} \frac{\exp(\gamma \nu_0) - 1}{\exp(\gamma \nu) - 1} \end{aligned} \quad (5)$$

where $T_d = 21 \text{ K}$, $A_d = 100 \mu\text{K}$, $\beta_d = 1$, $\nu_0 = 545 \text{ GHz}$.

- **Thermal SZ effect** occurs when CMB photons scatter off hot electrons in galaxy clusters. This inverse-Compton scattering alters the intensity of the cosmic background radiation in the direction of a cluster, resulting in a decrease of low-energy photons and an increase of higher-energy photons, particularly around 220 GHz. As a result, the spectrum no longer follows a perfect blackbody curve, but instead experiences a distortion:

$$s_{\text{sz}} = 10^6 y_{\text{sz}} / g(\nu) T_{\text{CMB}} \left(\frac{x(\exp(x) + 1)}{\exp(x) - 1} - 4 \right) \quad (6)$$

where $y_{\text{sz}} = 10^{-6}$ is the Compton parameter.

Note: The negative value of the thermal Sunyaev-Zeldovich (tSZ) effect below 220 GHz arises because the scattering of CMB photons by hot electrons reduces the number of low-energy photons, causing a decrement in intensity. Above 220 GHz, the photons are upscattered to higher energies, leading to an increment in intensity, creating the characteristic spectral signature of the tSZ effect.

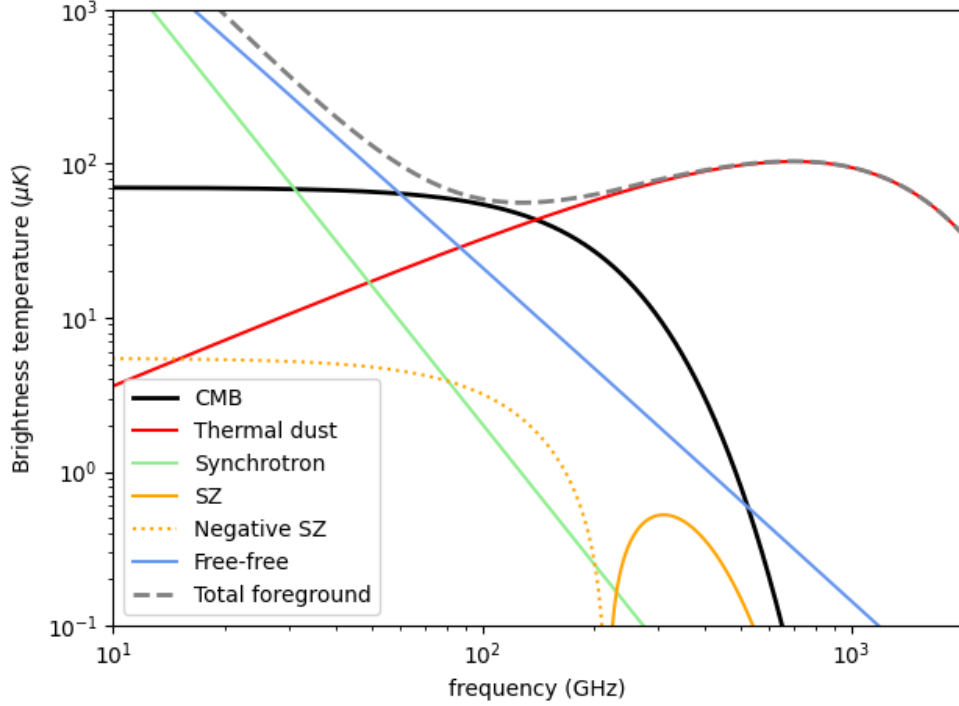


Figure 2: Spectral energy densities for the main astrophysical components

4 Planck temperature maps

Sky maps from the Planck PR4 data at various frequencies (100, 143, 217, 353, 545, and 857 GHz) are stored in HEALPix (Hierarchical Equal Area isoLatitude Pixelization) format. These maps are in FITS format, where each map is represented as a vector, and the HEALPix format allows for efficient representation of full-sky maps with uniform pixel coverage. The maps can be processed using the `healpy` package in Python, with chosen projection of `mollview` and ring scheme for easy analysis and visualization. Using function `read_map` and `mollview`, 6 raw maps from Planck can be visualized in figure 3.

It is obvious that there is dipole distortion cause by the motion of Solar System with respect to the CMB. We can read the dipole map provided by the Planck Collaboration and subtract it from our data. Following the section 2.3, the angular resolution differs across the various frequency channels. To ensure consistency, we need to convolve each map with a Gaussian beam so that all maps have the same resolution. The target Full Width at Half Maximum (FWHM) should be at least as high as the highest FWHM among the frequency channels; for this analysis, we choose a target resolution of $\text{FWHM} = 10$ arcmin. If the map x already has a resolution characterized by FWHM_x , we can compute the additional smoothing required to reach the target resolution using the following equation:

$$\text{FWHM}_{\text{rx}} = \sqrt{\text{FWHM}_t^2 - \text{FWHM}_x^2} \quad (7)$$

The quantity FWHM_{rx} represents the FWHM of the Gaussian beam that needs to be applied to the map x , ensuring that all frequency channels are smoothed to a consistent target

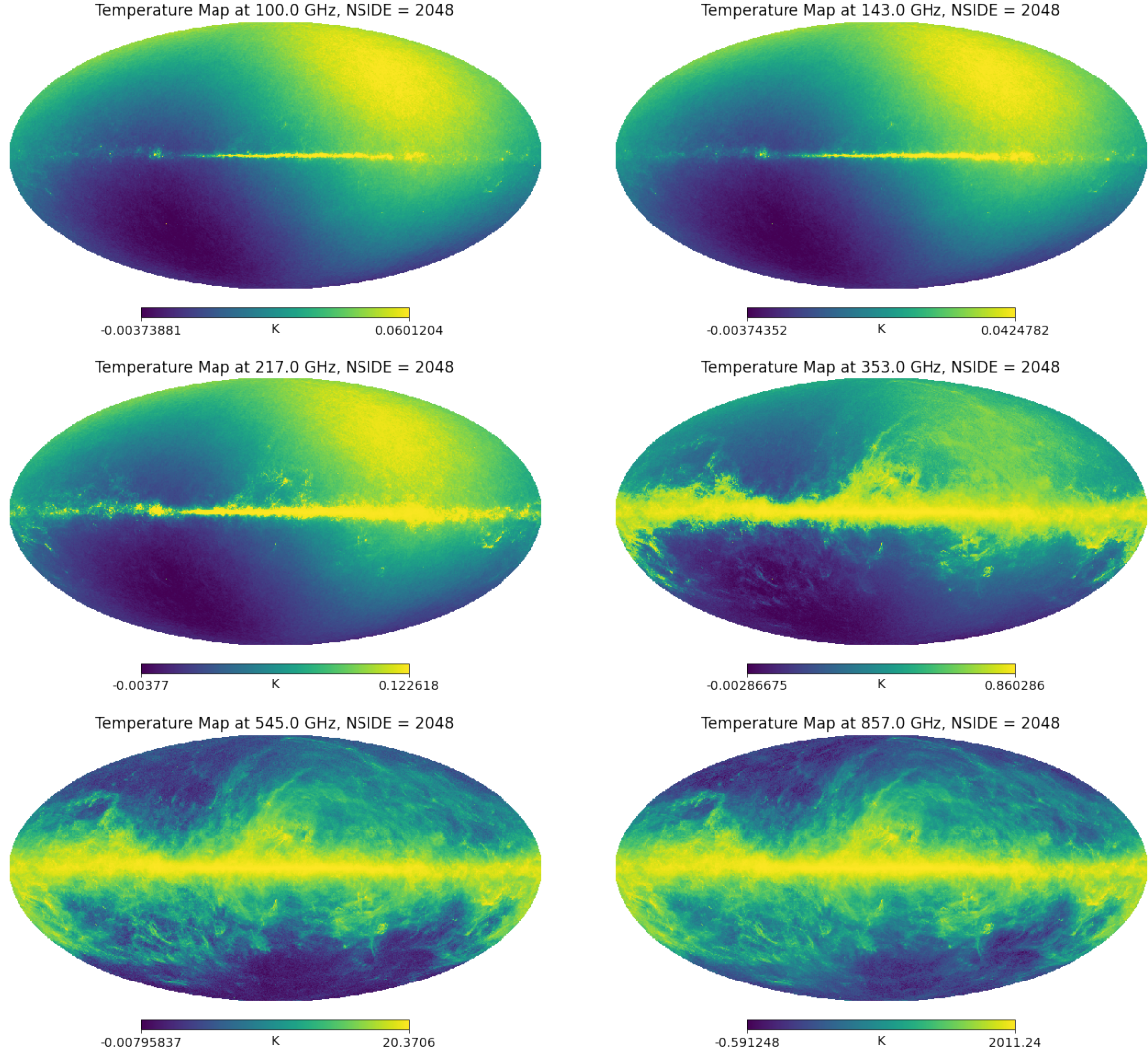


Figure 3: Planck Temperature Maps at 6 frequencies: 100, 143, 217, 353, 545, 857 GHz with NSIDE 2048

resolution. For this task, we will use the `smoothing` function from the `healpy` library.

Since a high NSIDE of 2048 is computationally demanding, we will reduce the pixel resolution by downgrading to an NSIDE of 512. This reduction is analogous to averaging four neighboring pixels to create a new one. To perform this operation efficiently, we will utilize the `up_grade` function from `healpy`.

The result of these treatments is shown in Figure 4. We can see that after the procedure, the map is much more homogeneous and isotropic, except for the Galactic plane in the middle of the sky map. Another notable observation is that as the frequency increases, the area and intensity of the Galactic emission also increase, showing that higher-frequency channels are more sensitive to the foreground emission from the Milky Way, which dominates at those wavelengths.

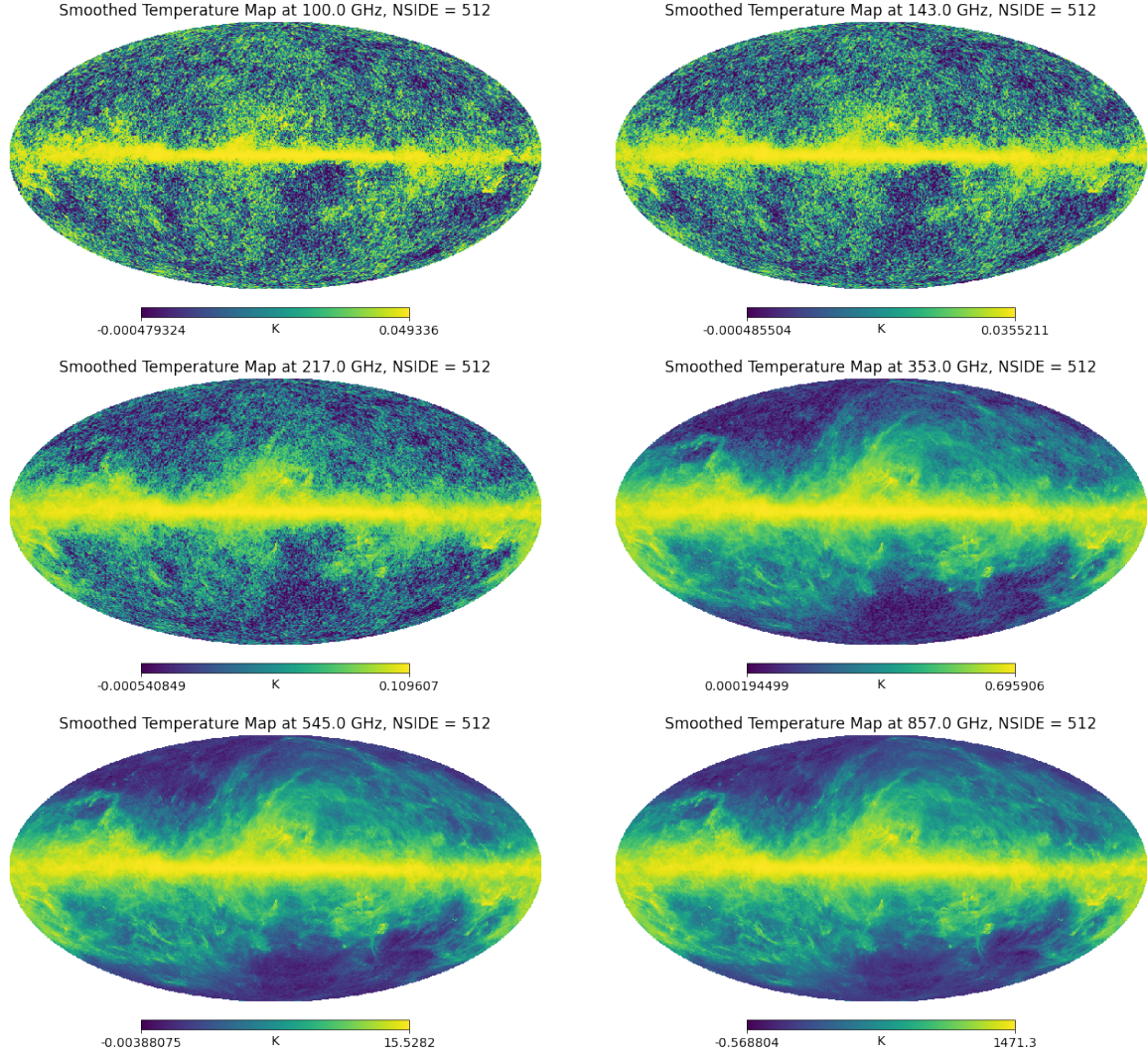


Figure 4: Planck Temperature Maps at 6 frequencies: 100, 143, 217, 353, 545, 857 GHz after removing dipole, smoothing to FWHM of 10 arcmin and downgrade to NSIDE of 512

5 Component Separation with MILCA

Having six frequency maps, we will apply the component separation method known as the Internal Linear Combination (ILC) to extract the CMB signal, following MILCA by [Hurier et al. \(2013\)](#).

The observed sky model is:

$$\mathbf{T}(\mathbf{p}) = \mathcal{A}\mathbf{S}(\mathbf{p}) + \mathbf{N}(\mathbf{p}) \quad (8)$$

where:

- $\mathbf{T}$ is the total observed map,
- $\mathbf{N}$ is the noise,

- $\mathbf{S}$ is the matrix of astrophysical components,
- $\mathcal{A}$ is the mixing matrix, describing how each component contributes to each observed channel.

ILC Reconstruction

The Internal Linear Combination (ILC) reconstructs the component S_c as a weighted sum of the observed maps:

$$\hat{S}_c(\mathbf{p}) = \sum_i w_i T_i(\mathbf{p}) \quad (9)$$

The weights $\mathbf{w} = (w_1, \dots, w_{N_{\text{obs}}})$ minimize the variance:

$$\text{Var}(\hat{S}_c) = \mathbf{w}^T \mathcal{C}_T \mathbf{w} \quad (10)$$

where $\mathcal{C}_T = \langle \mathbf{T} \mathbf{T}^T \rangle$ is the covariance matrix of the observational data.

Constraints

For CMB reconstruction, the constraint is:

$$g = \sum_i w_i = 1 \quad (11)$$

For other components, the constraint is:

$$g = \mathbf{w}^T \mathbf{f}_c = 1 \quad (12)$$

where $\mathbf{f}_c$ is the emission spectrum of the component.

Linear solution for component S_c

The weights are computed by solving the linear system:

$$\begin{pmatrix} 2\mathcal{C}_T & -\mathbf{f}_c \\ \mathbf{f}_c^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{w} \\ \lambda \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix} \quad (13)$$

This gives the optimal weights:

$$\hat{\mathbf{w}} = \mathcal{C}_T^{-1} \mathbf{f}_c (\mathbf{f}_c^T \mathcal{C}_T^{-1} \mathbf{f}_c)^{-1} \quad (14)$$

The reconstructed component is:

$$\hat{\mathbf{S}}_c(\mathbf{p}) = (\mathbf{f}_c^T \mathcal{C}_T^{-1} \mathbf{f}_c)^{-1} \mathbf{f}_c^T \mathcal{C}_T^{-1} \mathbf{T}(\mathbf{p}) \quad (15)$$

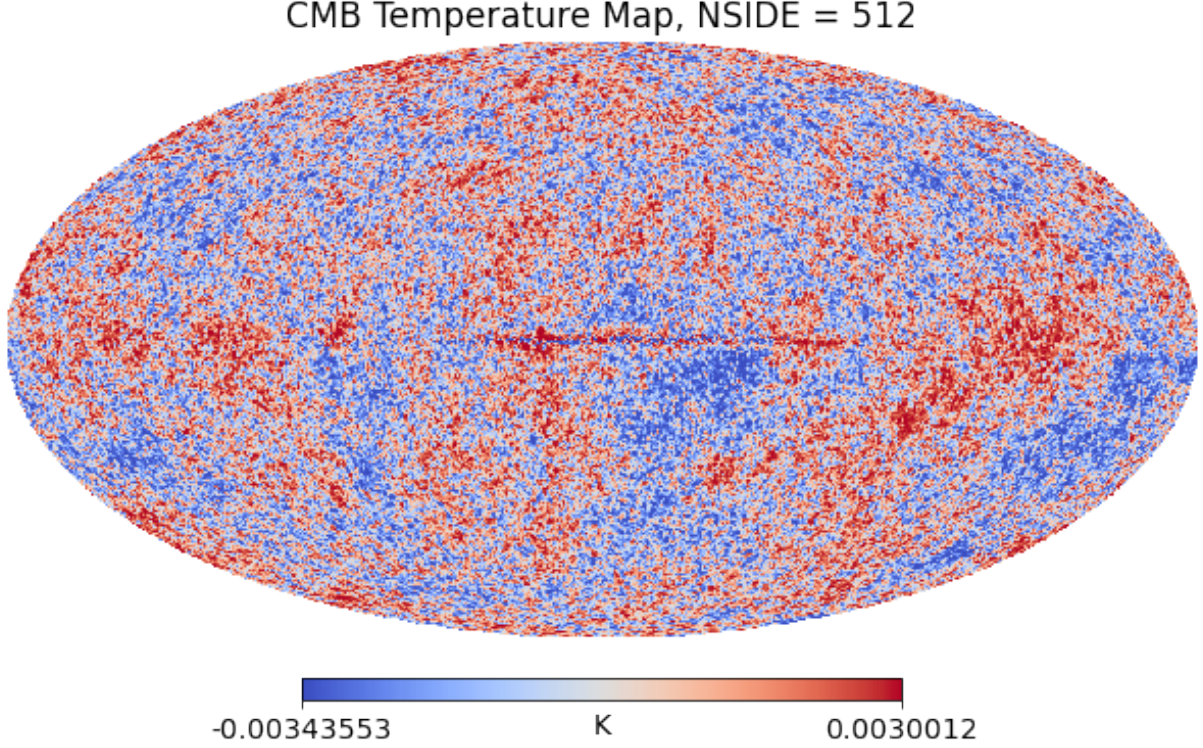


Figure 5: CMB Temperature Anisotropies with NSIDE of 512 obtained using MILCA

For CMB:

$$\widehat{\mathbf{S}}_c(\mathbf{p}) = \frac{\mathbf{1}^T \mathcal{C}_T^{-1} \mathbf{T}(\mathbf{p})}{\mathbf{1}^T \mathcal{C}_T^{-1} \mathbf{1}} \quad (16)$$

Using equation 16, with $\mathbf{T}(\mathbf{p})$ being the observational data (6 temperature maps), the result of MILCA for CMB is shown in figure 5. We can see that result is very similar to the map obtained by Planck Collaboration. However, we can still see a little bit disruption in the middle of galactic plane, where the galactic emissions dominate.

5.1 Masking the galactic plane

To reduce the signal from galactic plane and improve the statistics of CMB, we can define a mask of 10% pixel size by choosing a suitable threshold on the map with highest contamination at 857 GHz and apply it for all the channels (see figure 6). To compute the covariance with a masked map, we first utilize the `MaskedArray` feature provided by Python. Once we have the covariance, we can input it into equation 16 to compute the estimate of the signal, $\widehat{\mathbf{S}}_c(\mathbf{p})$. At this stage, we have the option to calculate $\widehat{\mathbf{S}}_c(\mathbf{p})$ using either the unmasked or the masked observational maps, $\mathbf{T}(\mathbf{p})$. The results of these computations are presented in Figure 7.

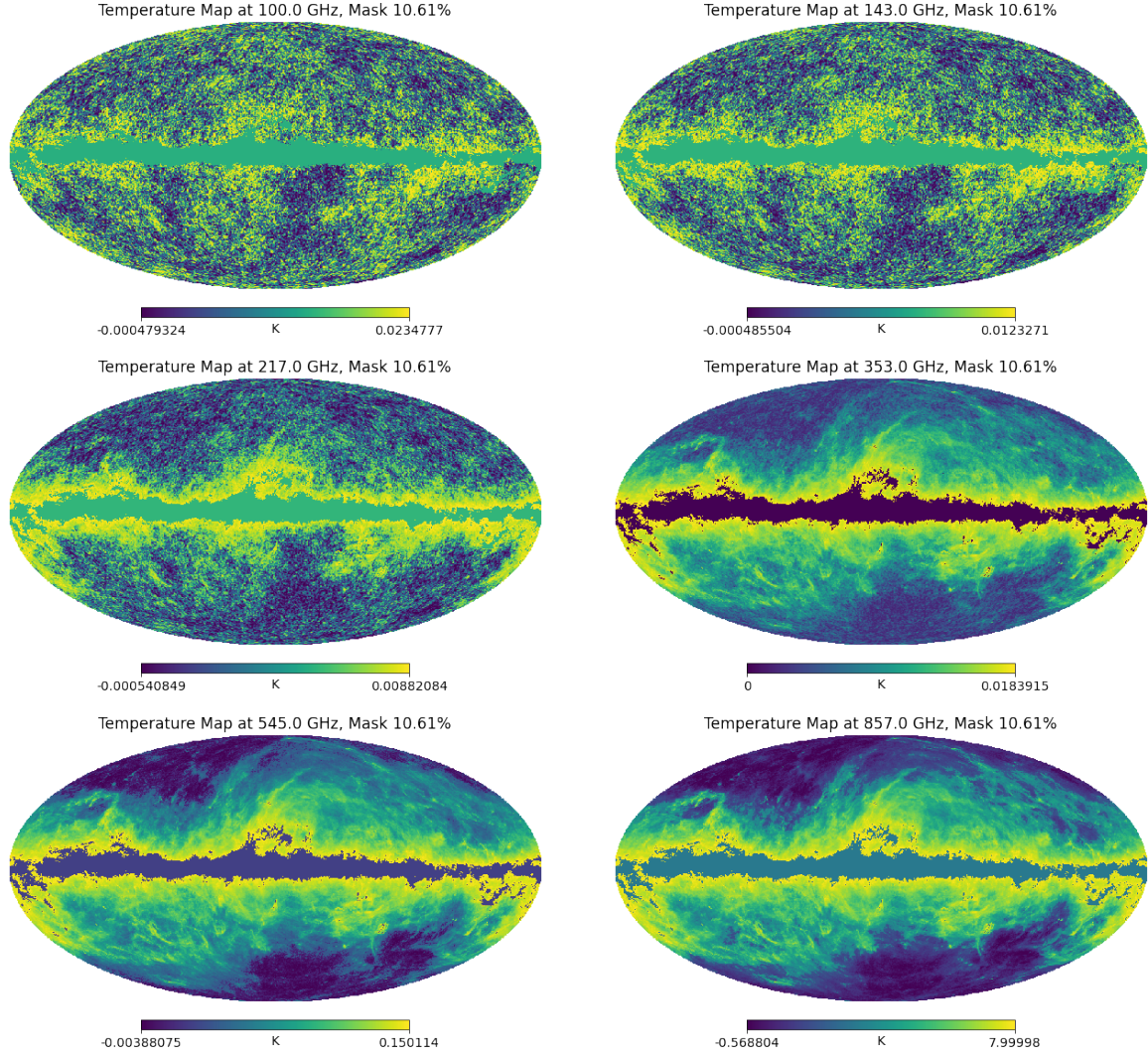


Figure 6: Planck temperature map with mask of 10% of pixel size

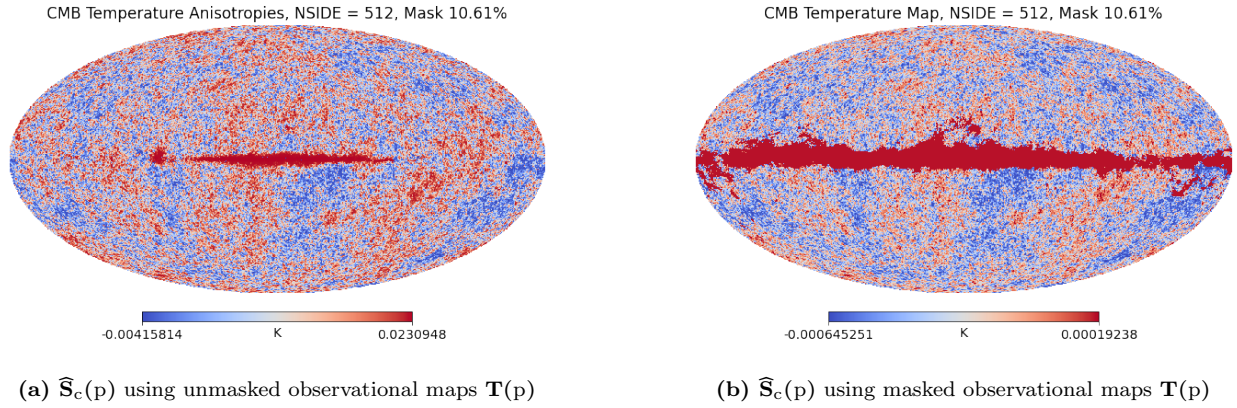


Figure 7: The CMB temperature map with MILCA using computed by covariance from masked maps

5.2 Attempts with NSIDE of 2048

I also attempted the same procedure using an NSIDE value of 2048. When using the unmasked maps, the result (figure 8) was very similar to the previous map of NSIDE of 512. However, when I tried to run the algorithm on the masked maps, my kernel failed. I suspect that the issue is related to memory limitations. As the number of pixels increases with higher NSIDE values (from 512 to 2048), the memory requirements grow significantly, particularly during the computation of the covariance matrix. The larger data set might exceed the available memory, leading to a kernel failure. This could be due to the increased size of the covariance matrix or the overall data, which could overwhelm the system's resources.

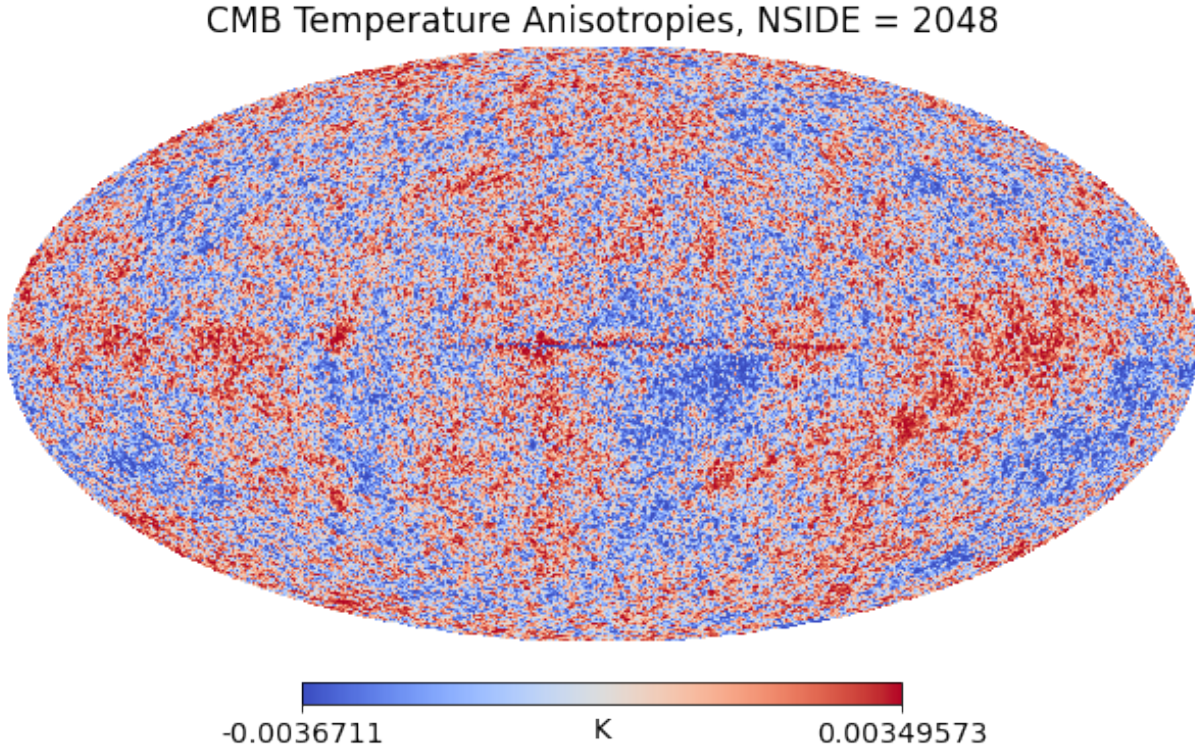


Figure 8: CMB Temperature map with MILCA of NSIDE 2048

5.3 Map of the thermal Sunyaev-Zeldovich effect

The Sunyaev-Zeldovich (SZ) effect is the phenomenon where cosmic microwave background (CMB) photons are scattered by hot electrons in galaxy clusters, making it a valuable tool for mapping and studying these clusters. Utilizing the general equation 15 of the Internal Linear Combination (ILC) method with masked maps from previous sections, we chose the conversion factors $\mathbf{f}_c = (-0.248 \ -0.359 \ 5.12 \ 0.161 \ 0.069 \ 0.038)^T$. This approach enabled us to generate the Sunyaev-Zeldovich map displayed in figure 9. In this map, the positive sources, represented by red dots, correspond to galaxy clusters, with larger dots potentially indicating superclusters. In contrast, the negative sources, marked as blue dots

Sunyaev-Zeldovich Map, NSIDE = 512, Mask 13.98%

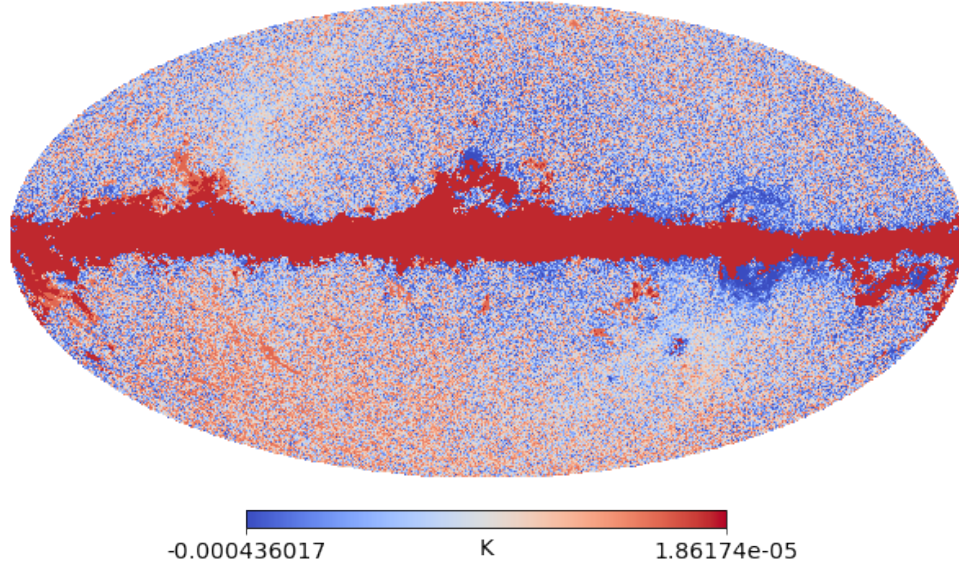


Figure 9: Sunyaev Zeldovich map

near our mask, represent strong galactic and extragalactic radio sources, which can act as contaminants in our analysis. These contaminants should ideally be masked or mitigated to improve the precision of our results.

To gain deeper insights into these clusters, we can apply gnomonic projection, which allows us to zoom in on specific areas for detailed study, as illustrated in figure 10. This approach allows for a more thorough examination of the properties and dynamics of the identified galaxy clusters.

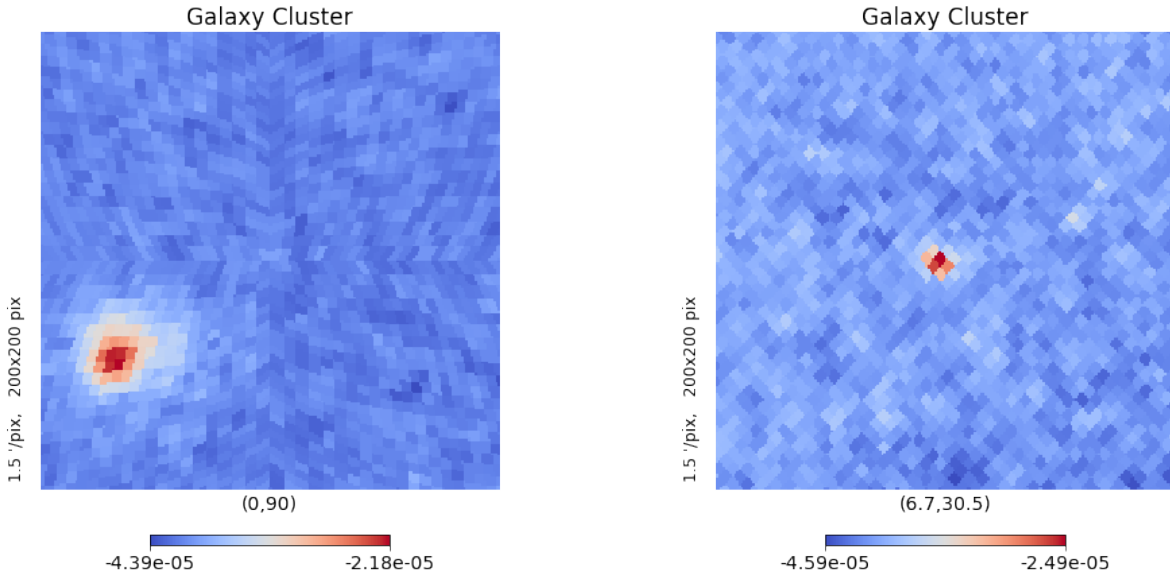


Figure 10: Gnomonic view of galaxy clusters

6 Angular Power Spectrum

The statistical properties of the CMB map, which show nearly Gaussian fluctuations, are predominantly described by the 2-point correlation functions. Assuming isotropy in the underlying probability distribution, the correlations between anisotropies at different points in the sky depend only on the angle between them. To analyze these correlations,

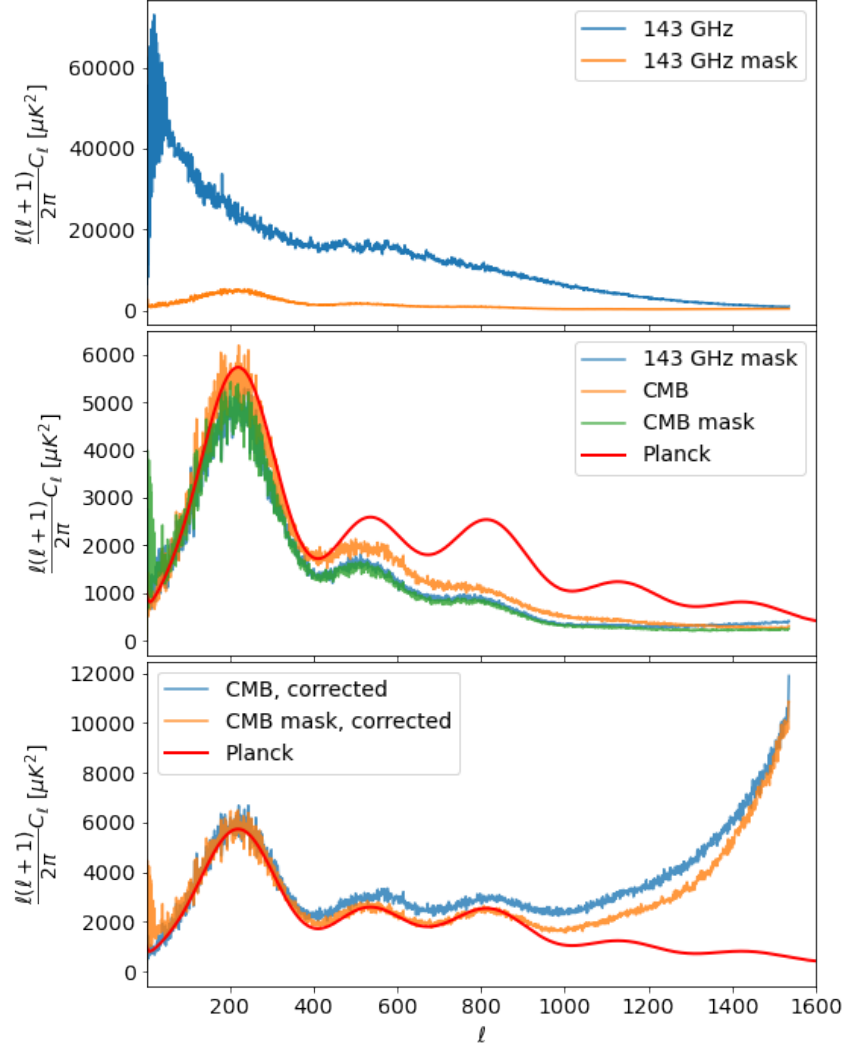


Figure 11: Angular power spectrum. *Top:* The power spectrum from Temperature map at 143 GHz with/without mask. *Middle:* The power spectrum from Temperature map of CMB with/without mask and the theoretical power spectrum fitted by Planck Collaboration. *Bottom:* The power spectrum of CMB with/without mask after applying the Gaussian beam and pixel-window correction.

temperature are typically expanded in terms of spherical harmonics:

$$\Delta T(\hat{n}) = \sum_{\ell,m} a_{\ell,m}^T Y_{\ell,m}(\hat{n}) \quad (17)$$

The index ℓ represents the degree, corresponding to the angular separation $\theta \sim \pi\ell$ in the map, while index m represents the order linked to orientation.

We can form the angular power spectra from the expansion coefficients $a_{\ell,m}^T$ as:

$$C_\ell^{TT} = \frac{1}{2\ell + 1} \sum_{\ell,m} a_{\ell,m}^T a_{\ell,m}^{T*} \quad (18)$$

This procedure can be performed using the `anafast` feature of `healpy`. As shown in Figure 11, for the map at 143 GHz, the unmasked map displays a dominant signal compared to the masked map, particularly at low frequencies, due to the contribution from the Galactic plane. Regarding the CMB maps, we observe two key points: (1) there is a noticeable damping of the second and third peaks compared to the best theoretical fit from Planck, and (2) the mask has reduced the overall amplitude of the power spectrum due to the decrease in the total number of pixels. This issue arises from the application of Gaussian smoothing and pixel averaging.

To address the effects of smoothing, we need to correct the power spectrum using the instrument beam. This can be achieved by using the `gauss_beam` feature of `healpy` to compute the transfer function of the beam, commonly referred to as B_ℓ . We then divide the amplitude of the power spectrum by B_ℓ^2 to account for this correction.

For pixel averaging, we employ the `pixwin` feature to obtain the pixel window function for an NSIDE of 2048. Similarly to the beam correction, we also divide the power spectrum by this pixel-window function to compensate for the effects of pixelization.

In the last sub-figure of figure 11, we see that the corrected power spectrum aligns well with the Planck results up to $\ell \sim 900$. However, beyond this point, the power spectrum is increasingly dominated by noise. The noise observed beyond $\ell \sim 900$ is probably a combination of instrumental noise and residual foreground contamination.

7 Cosmological Parameter Constraints

To derive constraints on cosmological parameters, we focus on fitting the angular power spectrum C_ℓ^{TT} up to multipole $\ell \sim 800$. This range is selected to minimize contamination from noise. For simplicity, we fit two parameters, A_s and n_s , using the theoretical curve generated from `CAMB`, a Python package for cosmological calculations.

The parameter A_s represents the amplitude of primordial scalar perturbations, which quantifies the initial fluctuations in the early universe. The parameter n_s , known as the scalar spectral index, describes the scale dependence of these fluctuations and indicates whether the primordial power spectrum is tilted relative to a scale-invariant spectrum ($n_s = 1$).

To fit the parameters A_s and n_s , we minimize the statistic χ^2 :

$$\chi^2 = \sum_{\ell=2}^{\ell_{\max}} \frac{(C_\ell^{\text{obs}} - C_\ell^{\text{model}})^2}{\sigma_\ell^2}, \quad (19)$$

where C_ℓ^{model} is computed using the theoretical expression for C_ℓ^{TT} and σ_ℓ is the uncertainty in the observed power spectrum, approximated as $\sigma_\ell = \frac{1}{\ell+1}$.

The fitting is performed using the Markov Chain Monte Carlo (MCMC) method implemented with `emcee`, which employs the Metropolis-Hastings algorithm. For this run, we used 10 walkers and 1000 steps, removing 50% of the burn-in phase. The results are shown in Figure 12. The contours exhibit Gaussian distributions as expected, though the contours are not as smooth as desired. This might improve by increasing the number of walkers or steps if more computational time is available.

The best-fit parameters obtained from the MCMC analysis are then used to plot the theoretical curve. Comparing this curve with the observational angular power spectrum, we find that the fit is quite good, especially for the first and second peaks.

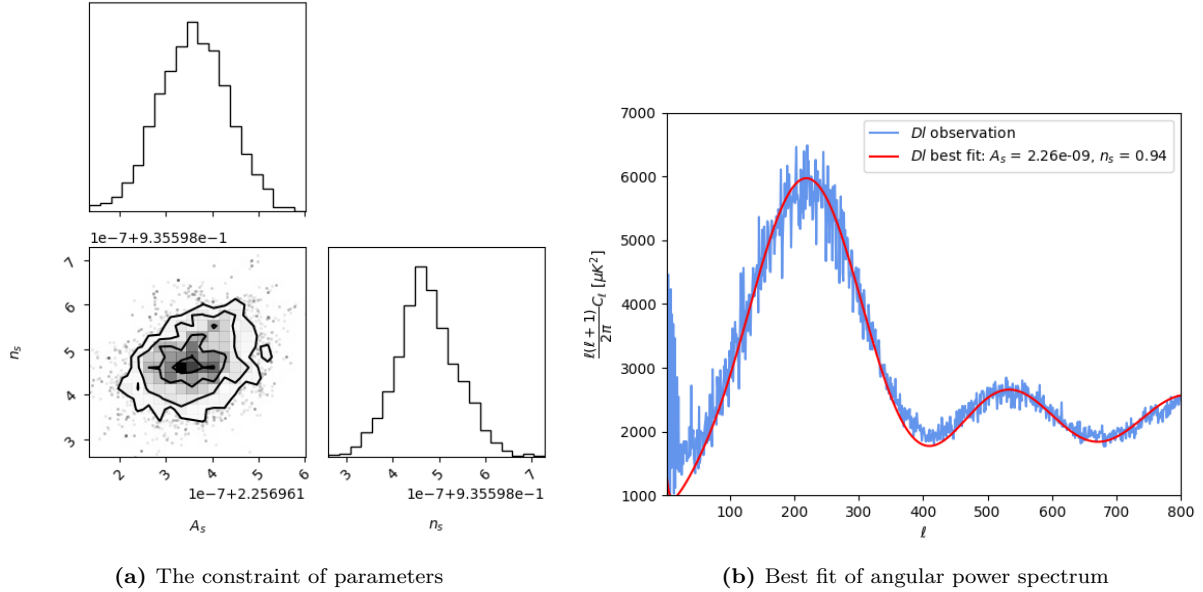


Figure 12: Results of MCMC parameter constraints and chi-squared fitting of the angular power spectrum.

References

- Draine, B. T. 2011, *Physics of the Interstellar and Intergalactic Medium*
- Fixsen, D. J. 2009, *The Astrophysical Journal*, 707, 916–920
- Hurier, G., Macías-Pérez, J. F., & Hildebrandt, S. 2013, *Astronomy and Astrophysics*, 558, A118
- Planck Collaboration X. 2016, *Astronomy and Astrophysics*, 594, A10
- Smoot, G. F., Bennett, C. L., Kogut, A., et al. 1992, , 396, L1